Neerav Chaudhary

APPLIED AI ENGINEER | PRODUCT SYSTEMS, LLM EVALUATION, HEALTHCARE AI

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LinkedIn | GitHub | Portfolio

Applied AI engineer and published researcher who takes ambiguous workflows from product direction and interface design through full-stack implementation and measurable evaluation. Builds voice agents, grounded research tools, healthcare safety systems, and realtime products with explicit permissions, source traceability, confirmation-gated actions, and evaluation harnesses.

SELECTED AI PRODUCTS

Relay - Voice-first employee service agent | React, TypeScript, Fastify, Deepgram

- Designed and built a voice and text assistant that grounds answers in employee records, navigates the portal, and executes support, reminder, meeting, and task workflows only after user confirmation.
- Built an evaluation harness across seven multi-turn workflow families: 103/110 development conversations and 39/42 held-out rewordings completed, with zero unsafe state changes across 21 adversarial runs.
- Implemented server-owned identity, role-scoped tools, idempotency, append-only audit records, and visible verification of final application state.

TraceLens - Healthcare AI safety and evaluation | Python, PyTorch, Hugging Face

- Built a preregistered workspace to measure emergency guidance using physician-labelled cases, hash-pinned models, data, and prompts, frozen analysis code, multi-GPU runners, and fail-closed imports.
- Trained Vega, an independent message detector (held-out AUC 0.966). It caught 18 of the chatbot's 31 missed emergencies, disclosed each flag, and raised final emergency guidance from 39/70 to 47/70 with one added false alarm.
- Developed a validated recommendation reader, support-loop replay, trajectory evaluator, blinded review desk, decision-model training package, and 300-test Python suite.

Refract - Source-grounded research workspace | React, FastAPI, PostgreSQL, ChromaDB

- Built five connected workspaces for close reading, passage-linked RAG, cross-paper synthesis, recall-first learning, dataset auditing, and baseline modeling inside one persisted research session.
- Added review gates that stop analysis at data-quality and leakage issues and preserve provenance from source passage to claim and accepted model result.

PROFESSIONAL EXPERIENCE

Deep Strat | Artificial Intelligence Research Intern **Aug-Nov 2023**

- Built a three-stage LLM workflow that classified customer complaints, retrieved routing context, and generated SOP-grounded cases; reduced churn-model false positives 25% while maintaining stable AUC.
- Advised founders and policy researchers on AI safety, guardrails, and failure modes; authored a 50-page technical reference that supported jointly authored AI-safety research presented at Nullcon.

Manipal University Jaipur | AI Researcher, Medical Imaging **Aug-Dec 2022**

- Co-authored an IEEE paper and built its repeatable Python/OpenCV pipeline for MRI preprocessing and brain-atrophy quantification across 2,198 images; anatomical cropping raised reported accuracy from 66% to 71%.
- Built a separate four-class TensorFlow experiment on 6,400 MRI images; probability-level ensembling improved held-out accuracy from 78% to 85%.

EDUCATION AND PUBLICATION

San Jose State University | M.S., Industrial and Engineering Management | GPA 3.81 **2024-2026**
Manipal University Jaipur | B.Tech., Computer Engineering | GPA 8.8/10 **2018-2022**

Publication: N. Chaudhary and S. S. Verma, "Analyzing Brain Atrophy in Dementia Using Computer Vision," IEEE ICRITO, 2022. doi:10.1109/ICRITO56286.2022.9964994

SKILLS

AI systems: LLM evaluation, harness engineering, voice agents, structured tool calling, RAG, multi-agent workflows, grounding, human-in-the-loop controls, Hugging Face, Deepgram, LangChain, OpenAI, Anthropic

Engineering: Python, TypeScript, React, FastAPI, Fastify, Express, SQL, PostgreSQL, SQLite, REST APIs, authentication and authorization, Docker, AWS, Git

ML and research: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, CNNs, Random Forest, hypothesis testing, calibration, data-quality and leakage audits, technical writing